

ECE 229 Final Project

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Abstract

This project is a re-implementation of [1], which considers the control of planar rotations. The problem of stabilizing a point on a unit circle will be considered, as it is equivalent to stabilizing a point in the group of planar rotations, $SO(2)$. As the paper points out, the control of orientation is non-trivial due to the fact that for systems where the state space is restricted to a compact set, there will be more than one globally asymptotically stable (GAS) equilibrium point.

This creates an issue where the selected control law could vanish in locations other than the desired equilibrium point. The goal of this project is to design a robust hybrid switching control law that uniquely renders a desired equilibrium point GAS, and to confirm the results through simulation.

1 Problem Statement

The hybrid system $\mathcal{H}$ is written as:

$$\mathcal{H} = \begin{cases} \dot{x} \in F(x), & x \in C, \\ x^+ \in G(x), & x \in D. \end{cases} \quad (1)$$

where:

- $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is the flow map of x .
- $C \subset \mathbb{R}^n$ is the flow set of x .
- $G : \mathbb{R}^n \rightrightarrows \mathbb{R}^n$ is the jump map of x .
- $D \subset \mathbb{R}^n$ is the jump set of x .

A generalization of the hybrid system that will be used can be formulated as:

$$\begin{aligned} \dot{x} &= f(x, \omega(x, \xi)) & x^+ &= x \\ \dot{\xi} &= \mathcal{F}(x, \xi), \underbrace{(x, \xi) \in \mathcal{C}}_{\text{Flow condition}} & \xi^+ &\in \mathcal{G}(x, \xi), \underbrace{(x, \xi) \in \mathcal{D}}_{\text{Jump condition}} \end{aligned} \quad (2)$$

The unit circle and planar rotations are denoted as:

$$\begin{aligned} \mathcal{S}^1 &= \{x \in \mathbb{R}^2 : x^\top x = 1\}, \\ SO(2) &= \{R \in \mathbb{R}^{2 \times 2} : R^\top R = I, \det(R) = 1\}. \end{aligned} \quad (3)$$

The dynamics for system A are modeled as:

$$\dot{x} = \omega Sx \quad (4)$$

where

$$S = \mathcal{R} \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \quad (5)$$

and ω is the angular velocity and input for the system.

A goal is to a design switching feedback control law such that the set $\mathcal{A} = \{e_1\} \times \mathcal{X}$ is GAS for the closed-loop system A, where $\mathcal{X} \subset \mathbb{R}^n$ is compact and $e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

The dynamics of the system B where the input is angular acceleration are modeled as:

$$\begin{aligned}\dot{x} &= \omega Sx \\ \dot{\omega} &= u\end{aligned}\tag{6}$$

The ultimate goal is to use the result for system A to design a switching feedback control law such that the set $\mathcal{B} = \{e_1\} \times \{0\} \times \mathcal{X}$ is GAS for the closed-loop system B, using the same flow set, jump set, and jump map as used for system A.

2 Analysis

2.1 Transformation of Maxima for a Lyapunov Function

Initially, one might consider the following potential function as a feasible Lyapunov function candidate for system A:

$$P^*(x) = \frac{1}{2}(1 - x_1).\tag{7}$$

As proven in Example 3.2 in [1], there are two points for which

$$\langle \nabla P^*(x), Sx \rangle = 0.$$

These points are $x = e_1$ which is the minimum of $P^*(x)$, and $x = -e_1$ which is the maximum of $P^*(x)$. If one were to design a control law using a feedback based on the Lyapunov function $P^*(x)$, an approach could be to switch the signs of the feedback based on the sign of x_2 . This would render the point e_1 GAS for every x except for $-e_1$, where the control law would vanish. Therefore, if the initial condition of the system is at $-e_1$ or small perturbations push x into $-e_1$, the solution would be maintained there indefinitely.

For a more feasible approach, it would be ideal to keep the minimum at e_1 to keep the point GAS, but to transform the maximum. In order to prevent propagating the issue to a different equilibrium point, it would be necessary to use at least two different Lyapunov functions for which the maxima are different, such that if the solution of H is close to the maximum of one of the Lyapunov functions, the feedback is switched such that the solution follows the gradient of a decreasing Lyapunov function.

This transformation of the maximum while preserving the minimum can be done using diffeomorphisms of S^1 . This rotates S^1 while keeping e_1 fixed, for which the amount of rotation applied can be determined by a gain $k \in \mathbb{R}$.

The diffeomorphism $\mathcal{T}$ of S^1 is defined as

$$\mathcal{T}(x, k, P) = \Phi(x, k, P)x\tag{8}$$

where

$$\Phi(x, k, P) = \exp(kP(x)S).\tag{9}$$

and

$$\exp(\omega S) = \begin{bmatrix} \cos(\omega) & -\sin(\omega) \\ \sin(\omega) & \cos(\omega) \end{bmatrix}.\tag{10}$$

The bound for k where the diffeomorphism holds is:

$$|k| < \frac{1}{\max\{\|\nabla P(x)\| : x \in S^1\}}.\tag{11}$$

Considering $P(x) = P^*(x)$ from (7),

$$|\nabla P^*(x)| = 1/2.$$

Therefore, $|k| < 2$ is a valid selection of gain k .

2.2 Hybrid Feedback Control Law For System A

The hybrid control structure of system A introduces the switching variable $q \in \mathcal{Q}$, where $\mathcal{Q} = \{1, 2\}$. Furthermore, it introduces:

$$\begin{aligned} T_q(x) &= \mathcal{T}(x, k_q, P_q) \\ V_q(x) &= V(T_q(x)) \\ \mathcal{M}(x) &= \min\{V_q(x) : q \in \mathcal{Q}\}. \end{aligned} \tag{12}$$

The flow and jump sets are defined as:

$$\begin{aligned} C &= \{(x, q) \in \mathcal{S}^1 \times \mathcal{Q} : \mathcal{M}(x) - V_q(x) \geq -\delta\} \\ D &= \{(x, q) \in \mathcal{S}^1 \times \mathcal{Q} : \mathcal{M}(x) - V_q(x) \leq -\delta\}. \end{aligned} \tag{13}$$

where $\delta > 0$ can be selected as a hysteresis for switching between the feedback modes.

The jump map is defined as:

$$g(x) = \{q \in \mathcal{Q} : V_q(x) = \mathcal{M}(x)\}. \tag{14}$$

By Theorem 5.3 [1], the hybrid system $\mathcal{H}_1$ is well-posed and renders $\mathcal{A}$ GAS, and is defined as follows:

$$\begin{aligned} \dot{x} &= f(x, \omega) & x^+ &= x \\ \dot{q} &= 0, \quad (x, q) \in C & q^+ &\in g(x), \quad (x, q) \in D \end{aligned} \tag{15}$$

$\underbrace{\hspace{10em}}$
Flow condition
 $\underbrace{\hspace{10em}}$
Jump condition

Solving for the transformed Lyapunov functions for system A gives:

$$\begin{aligned} \mathcal{T}(x, k, P) &= \begin{bmatrix} \cos\left(\frac{k}{2}(1-x_1)\right) & -\sin\left(\frac{k}{2}(1-x_1)\right) \\ \sin\left(\frac{k}{2}(1-x_1)\right) & \cos\left(\frac{k}{2}(1-x_1)\right) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ \mathcal{T}(x, k, P) &= \begin{bmatrix} \cos\left(\frac{k}{2}(1-x_1)\right) x_1 - \sin\left(\frac{k}{2}(1-x_1)\right) x_2 \\ \sin\left(\frac{k}{2}(1-x_1)\right) x_1 + \cos\left(\frac{k}{2}(1-x_1)\right) x_2 \end{bmatrix} \\ V_q(\mathcal{T}(x, k, P)) &= \frac{1}{2} \left(1 - \cos\left(\frac{k}{2}(1-x_1)\right) x_1 + \sin\left(\frac{k}{2}(1-x_1)\right) x_2 \right) \end{aligned}$$

which implies:

$$\langle \nabla V_q(\mathcal{T}(x, k, P)), Sx \rangle \omega = \left(\frac{1}{4} (kx_2 + 2) \left(x_2 \cos\left(\frac{k}{2}(x_1 - 1)\right) - x_1 \sin\left(\frac{k}{2}(x_1 - 1)\right) \right) \right) \omega$$

Selecting $\omega = -\langle \nabla V_q(\mathcal{T}(x, k, P)), Sx \rangle$ trivially satisfies Assumption 3.3 [1], meaning that this is a control law that renders $\mathcal{A}$ asymptotically stable for system A, with a basin of attraction containing $\mathcal{S}^1 \setminus \{V_q^{-1}(1)\}$.

Now, in determining two different feedbacks, the gains k_1 and k_2 were selected to satisfy (11). These were selected to be $k_1 = 1.5$ and $k_2 = -1.5$, which gives the following Lyapunov potential functions.

$$V_1(\mathcal{T}(x, k, P)) = \frac{1}{2} \left(1 - \cos\left(\frac{1.5}{2}(1-x_1)\right) x_1 + \sin\left(\frac{1.5}{2}(1-x_1)\right) x_2 \right)$$

and

$$V_2(\mathcal{T}(x, k, P)) = \frac{1}{2} \left(1 - \cos\left(\frac{-1.5}{2}(1-x_1)\right) x_1 + \sin\left(\frac{-1.5}{2}(1-x_1)\right) x_2 \right)$$

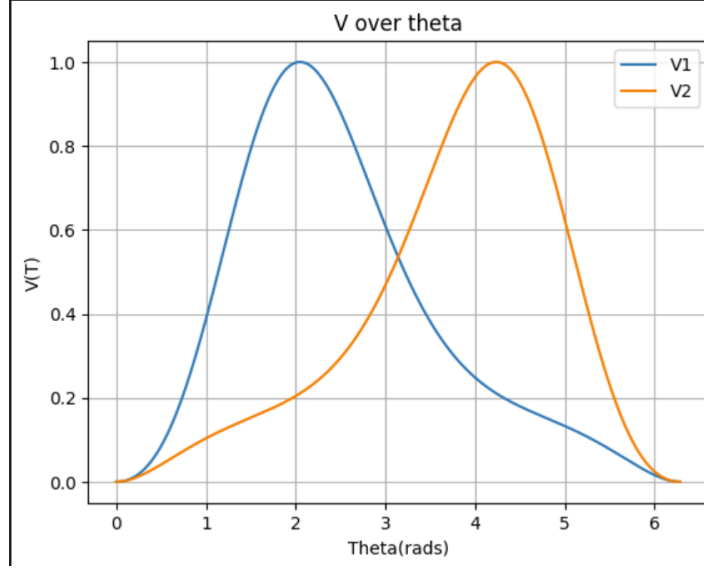


Figure 1: V1 and V2 plotted over θ

As can be seen in Figure 1, the functions have different maxima while maintaining e_1 as the common minimum. A value of $\delta = 0.25$ considering the region where flows and jumps are allowed, was chosen based on this plot. . The parameter δ can be interpreted as a region added to the curve defining the minimum of the two functions. The main idea is that once the state is outside the δ -region, the control law will switch to the other feedback for which the state will follow the surely decreasing gradient of the associated Lyapunov function. It is important that δ is chosen to be large enough to prevent infinitely many jumps at the minimum of the two functions, while being small enough to allow for jumping in local neighborhoods of the maxima.

2.3 Hybrid Feedback Control Law For System B

The proof of system A being GAS for $\mathcal{A}$ can be used to determine that the following system is also GAS for B , with the same flow set, jump set and jump map as for $\mathcal{H}_1$. The proof is supplemented with Theorem 8.8 and Theorem 7.12 in [2].

Considering the following Lyapunov function candidate:

$$\mathcal{W}(x) = V_q(\mathcal{T}) + \frac{1}{2}\omega^2 \quad (16)$$

As of Theorem 5.3 [1],

$$V_q((\mathcal{S}^1 \times Q) \setminus \mathcal{A}) > 0 \quad \text{and} \quad V_q(x) = 0 \text{ if and only if } x \in \mathcal{A}.$$

which implies that

$$\mathcal{W}(x)((\mathcal{S}^1 \times Q) \setminus \mathcal{A}) > 0 \quad \text{and} \quad \mathcal{W}(x) = 0 \text{ if and only if } \xi \in \mathcal{A}.$$

Next,

$$\langle \nabla \mathcal{W}(x), Sx\omega \rangle = \langle \nabla V_q(\mathcal{T}), Sx \rangle \omega + \omega u$$

Selecting $u = -\langle \nabla V_q(\mathcal{T}), Sx \rangle - \omega$.

$$\langle \nabla \mathcal{W}(x), Sx\omega \rangle = -\omega^2 \leq 0$$

The inequality only becomes an equality if $\omega = 0$, for which

$$\mathcal{W}(x) = V_q(\mathcal{T})$$

From the proof of Theorem 5.3 [1],

$$\langle \nabla V^*(\xi), F(\xi) \rangle \leq 0 \quad \text{for all } \xi \in \mathcal{S}^1 \times Q \supset C,$$

and

$$\langle \nabla V_q(\xi), F(\xi) \rangle < 0 \quad \text{for all } \xi \in C \setminus A.$$

This means that the only complete solutions for which V_q is equal to a constant for all times are when $\xi \in A$, for which $V_q(e_1) = 0$.

This proves that the only possible way complete solutions can keep $\mathcal{W}(x)$ equal to a constant is when that constant is 0. Therefore, there are no complete solutions that keep $\mathcal{W}(x)$ equal to a non-zero constant.

Finally, the proof of Theorem 5.3 [1] states:

$$\mathcal{M}(x) - V_q(x) \leq -\delta \tag{17}$$

which indicates

$$\mathcal{M}(x) - V_q(x) - \frac{1}{2}\omega^2 \leq -\delta.$$

Hence, there is still a strict decrease at jumps.

As can be seen, the additional $\frac{1}{2}\omega^2$ term does not interfere with any of the three Lyapunov function conditions. Since the term does not introduce any new equilibrium points, the switching logic remains dependent solely on $V_q(x)$. Therefore, the same flow set, jump set, and jump map determined for system A can be used to establish GAS of $\mathcal{B}$ for system B.

3 Simulations

The simulation of solutions of the hybrid control structure for system B supports the conclusions made from the analysis. The following three figures all confirm that $\mathcal{B}$ is GAS for the closed-loop system, but are simulated solutions from different initial conditions.

Figure 2 shows a simulation starting from (0,1), $\omega = 0$. In contrary to the approach taken with the Lyapunov function (7), this now provides smooth behavior at what used to be an unstable equilibrium point.

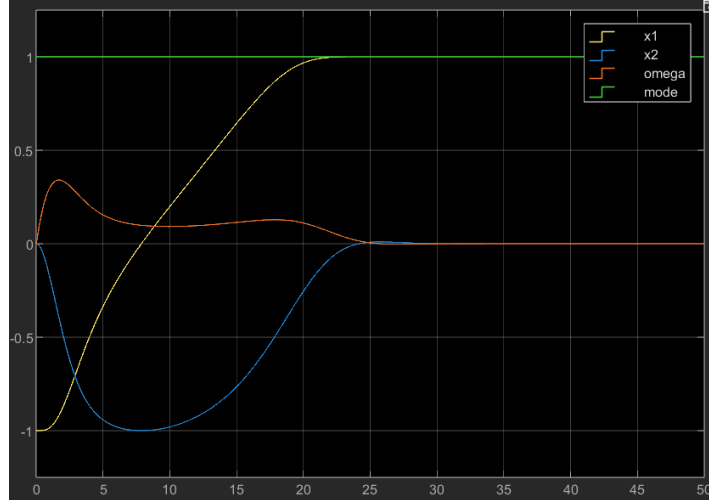


Figure 2: System B over time: Case 1

Figure 3 shows how the switching feedback operates when starting at the unstable maximum of V_1 with $\omega = 0$, at mode $q = 1$.

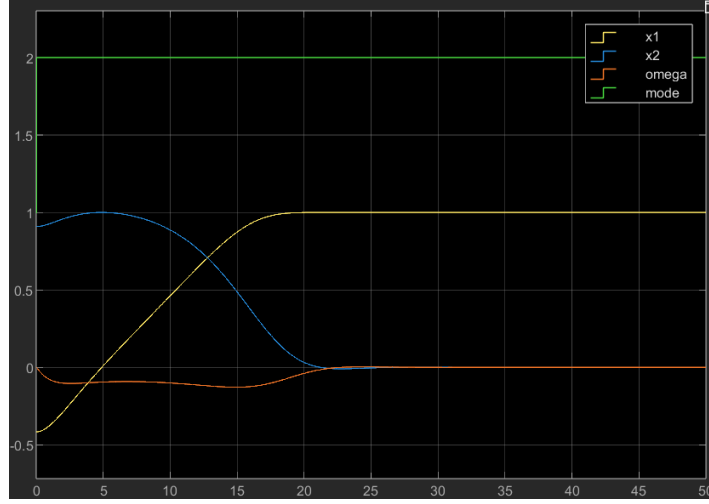


Figure 3: System B over time: Case 2

Figure 4 shows how the switching feedback operates when starting at the unstable maximum of V_2 with $\omega = 0$, at mode $q = 2$.

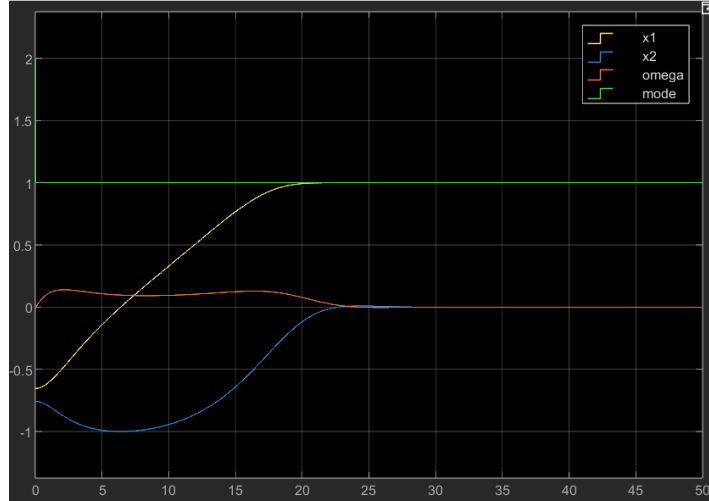


Figure 4: System B over time: Case 3

Although not truly reflected through the included figures, the maximum number of jumps for the system is 1.

4 Conclusion

In this project, [1] was re-implemented with the purpose of developing a hybrid control structure that renders a desired equilibrium point GAS robust to the inevitable presence of unstable equilibrium points for systems evolving on compact manifolds. This was done by transforming a Lyapunov function into two Lyapunov functions with different maxima, corresponding to two separate feedbacks. This was first designed for a system A with angular velocity as an input, before being extended to designing a hybrid switching feedback control law for system B with angular acceleration as the input, utilizing the same flow set, jump set, and jump map as for system A. Solutions of the hybrid control structure for system B were simulated in conditions that could be problematic in a non-hybrid approach, for which the system showcased desired behavior.

References

- [1] C. G. Mayhew and A. R. Teel, "Hybrid control of planar rotations," *Proceedings of the 2010 American Control Conference*, Baltimore, MD, USA, 2010.
- [2] R. G. Sanfelice, R. Goebel, and A. R. Teel, *Hybrid Dynamical Systems: Modeling, Stability, and Robustness*. Princeton, NJ, USA: Princeton Univ. Press, 2021.